

Weak enhancer allows auto-activation of *Irf8* to control cDC1 versus cDC2 lineage commitment

The *Irf8*+32-kb enhancer controls transcriptional autoactivation to generate 'IRF8 on' and 'IRF8 off' states that define type 1 conventional dendritic cell (cDC1) and cDC2 lineages, respectively. Weak affinity of this enhancer for BATF3–JUNB–IRF8 complexes allows specified cDC2 progenitors to successfully turn off *Irf8* transcription. If this enhancer is made stronger, cDC2 commitment and cDC1–cDC2 lineage segregation are compromised.

This is a summary of:

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The question

The lineage-determining transcription factor IRF8 enforces commitment to the type 1 conventional dendritic cell (cDC1) lineage through autoactivation, whereby IRF8 promotes its own transcription by binding to *cis*-regulatory sites near its gene. The cDC1 transcriptional program relies on high levels of IRF8, which cooperatively interacts with BATF3 and JUNB at motifs known as AP-1–IRF composite elements (AICEs) to control genes specifically expressed by cDC1s¹. The high level of IRF8 expression is maintained in cDC1s by autoactivation at an enhancer located 32 kilobases downstream of the *Irf8* transcriptional start site (the *Irf8*+32-kb enhancer), which contains several AICEs. Intriguingly, computational analysis and *in vitro* assays such as electrophoretic mobility-shift assays suggest that these AICEs have weaker binding affinities for BATF3–JUNB–IRF8 complexes than other AICEs engaged in cDC1s have. Yet without these weak AICEs, cDC1s fail to develop².

As they are arguably the most important AICEs for cDC1 development, we wondered why these elements are such poor binders for their regulatory factors, whether this means anything and what the consequences would be of improving their ability to interact with their factors.

The discovery

To answer these questions, we used CRISPR–Cas9 gene-editing technology to replace the weak AICEs in the native *Irf8*+32-kb enhancer with high-affinity AICEs and increase their number. This 'optimization' disrupted the development and function of both cDC1 and cDC2 lineages in mice.

First, we discovered that the commitment of pre-cDC2s to the cDC2 fate relies on a sub-optimized *Irf8*+32-kb enhancer. Pre-cDC2s express IRF8 but lack BATF3 expression. As they differentiate into cDC2s, IRF8 levels decrease and BATF3 levels increase³. During this dynamic process, the threshold for autoactivation at the native *Irf8*+32-kb enhancer is never met. With the optimized enhancer, however, many pre-cDC2s failed to downregulate IRF8 expression, becoming either cDC1s or a new hybrid DC subset with characteristics of both cDC1s and cDC2s (Fig. 1). Consequently, the number of cDC2s was reduced, which impaired their specific role in defending against helminths. Unexpectedly, we also found that low-affinity AICEs are required for the maximization of IRF8 expression in differentiated cDC1s. Mice with the optimized enhancer failed to mount cDC1-dependent CD8⁺ T cell

responses directed against tumors. Finally, our study revealed that suboptimization of the *Irf8*+32-kb enhancer is crucial for distinct transcriptional segregation of cDC1s versus cDC2s. In mice carrying the optimized enhancer, cDC1s upregulated several cDC2-associated genes, and a new hybrid DC subset with features of both lineages emerged.

The implications

Our findings reveal the crucial role of suboptimization of the *Irf8*+32-kb enhancer in cDC1 versus cDC2 lineage segregation. Established in other developmental processes, such as heart and limb development, enhancer suboptimization involves the strategic use of low-affinity motifs and suboptimal numbers and arrangements of motifs^{4,5}. This study extends the concept of enhancer suboptimization to hematopoiesis.

One limitation of this research is the inability to isolate which aspects of suboptimization, such as motif affinity and number, are critical for the integrity of cDC2 lineage commitment. Although cDC2 commitment was disrupted in mice with an optimized *Irf8*+32-kb enhancer, this enhancer variation increased not only the affinity but also the number of AICEs. Nonetheless, we observed that low affinity is a conserved feature of this enhancer, and our results from additional mouse lines and reporter assays demonstrate that any degree of enhancer optimization compromises *Irf8* expression and cDC1 versus cDC2 segregation. Another limitation of the current study is the lack of characterization of the chromatin landscape and transcription factor binding status at the optimized enhancer, which represents an area for future investigation.

On the basis of our observations of the *Irf8*+32-kb enhancer, we hypothesize that enhancer suboptimization is a general strategy of autoregulatory enhancers, particularly those that control lineage-determining transcription factors. We propose that such enhancers use suboptimal motifs and arrangements to establish a threshold for autoactivation. In this way, the enhancer is sensitive to varying transcription factor concentrations, with lower concentrations leading to an off state and one cell fate, and higher concentrations leading to an on state and another cell fate. The counterintuitive observation that higher-affinity AICEs reduced *Irf8* expression in cDC1s presents another intriguing avenue for future exploration.

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EXPERT OPINION

“Stronger isn't always better when it comes to developmental gene control: numerous studies show that enhancers bind transcription factors with sub-maximal affinity to restrict gene expression to the right cell types and developmental stages. This study adds an important aspect:

increased transcription factor binding not only drives ectopic gene expression but can also reduce enhancer strength and impair target gene expression.” **Matthias Merkenschlager, MRC Laboratory of Medical Sciences, London, UK.**

FIGURE

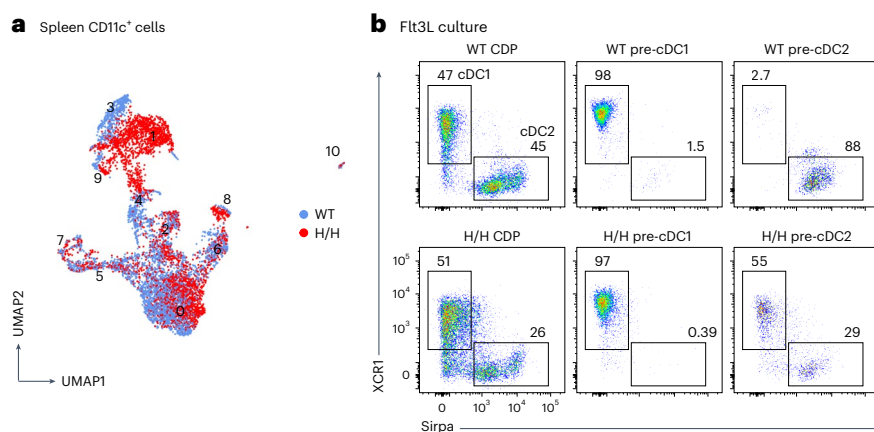


Fig. 1 | An optimized *Irf8*+32-kb enhancer disrupts cDC1 versus cDC2 lineage segregation. **a**, Splenic CD11c⁺ cells from mice with the native enhancer (WT) or the optimized *Irf8*+32-kb enhancer (H/H) were analyzed by single-cell RNA sequencing. Visualization by UMAP shows clusters of cDC1s (clusters 3 and 9) and cDC2s (clusters 0, 5 and 7). Cluster 1 represents the new hybrid DC population present only in mice with the H/H enhancer. **b**, DC progenitors, including common dendritic cell progenitors (CDP), pre-cDC1s and pre-cDC2s, from WT and H/H mice were cultured for 4 days in Flt3L, and their output cDC populations were analyzed by flow cytometry. H/H CDPs produced more cDC1s and fewer cDC2s than WT CDPs produced. H/H pre-cDC2s produced about 50% cDC1s, in contrast to the nearly complete absence of cDC1s from WT pre-cDC2s. © 2024, Ou, F. et al.

BEHIND THE PAPER

This work was inspired by our observation that sequences from the *Irf8*+32-kb enhancer did not bind BATF3–JUNB–IRF8 complexes in electrophoretic mobility-shift assays. We initially hypothesized that inserting multiple high-affinity AICEs into the enhancer would prevent specified cDC2 progenitors from downregulating IRF8 expression, potentially creating a cDC2-deficient mouse model. To our

disappointment, the H/H mice did not completely lack cDC2s. Nonetheless, these mice exhibited numerous problems beyond just reduced cDC2 numbers, which eventually led to unexpected discoveries. This work not only introduces the concept of enhancer suboptimization to immunology but also broadens the fundamental understanding of enhancer functionality. **F.O.**

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FROM THE EDITOR

“In this study, the authors find that enhancer suboptimization controls autoactivation of the gene encoding IRF8 to allow proper lineage divergence during cDC1 and cDC2 development.” **Stephanie Houston, Senior Editor, Nature Immunology.**